Homework (Lecture 10, slide 183, Time Series)

① $\ell_6$ : $\ell_6 = \alpha \frac{y_6}{S_2} + (1-\alpha)(\ell_5 - b_5)$

2006, Q2

$= 0.4406 \frac{29.3505}{0.7701} + (1-0.4406) \times (40.2696 - 0.7545)$

$= 39.7413$

② $b_{43}$ : $b_{43} = \beta(\ell_{43} - \ell_{42}) + (1-\beta) b_{42}$

2015, Q3

$= 0.0134(57.9607 - 63.9644) + 0.9866(0.7296)$

$= 6.6394$

③ $S_6$ : $\underset{Q2}{S_6} = 0.0023\left(\frac{y_6}{\ell_5 + b_5}\right) + (1-0.0023) \times \underset{Q2}{S_2}$

2006, Q2

$= 0.0023\left(\frac{29.3505}{40.2696 + 0.7545}\right) + (1-0.0023)\,0.7701$

$= 0.76997 \approx 0.7700$

④ $\hat{y}_{t+4|t} = (\ell_{44} + 4b_{44}) S_{44}$

$h=4$ ; last Q4

2016, Q4

$= (61.1697 + 4(0.6738))\,1.0254$

$= 65.4871$

more forecasted values :

a) $\hat{y}_{t+1|t} = (\ell_{44} + 1b_{44}) S_{41} \longrightarrow$ last Q1

$\overset{T}{44} + \overset{h}{1} - \overset{m}{4}$

$= (61.1697 + 0.6738)\,0.9701$

b) $\hat{y}_{t+2|t} = (\ell_{44} + 2b_{44}) S_{42} \longrightarrow$ last Q2

$\overset{T}{44} + \overset{h}{2} - \overset{m}{4}$

$= (61.1697 + 2(0.6738))\,0.7685$

$= 48.0445$

c) $\hat{y}_{t+3|t} = (\ell_{44} + 3b_{44}) S_{43} \longrightarrow$ last Q3

$\overset{T}{44} + \overset{h}{3} - \overset{m}{4}$

$= (61.1697 + 3(0.6738))\,1.2358$

$= 78.0916$

d) $\hat{y}_{t+5|t} = (\ell_{44} + 5b_{44}) S_{41} \longrightarrow$ last Q1

$44 + 3 - 4\left(\frac{5-1}{4} + 1\right)$ ; $k = \frac{h-1}{m}$

$= (61.1697 + 5(0.6738))\,0.9701$

$= 62.6090$ etc.

⇒ essentially "h" determines the difference between $\hat{y}_{t+1|t}$ and $\hat{y}_{t+5|t}$